$$(43 + 57) + 25 = 43 + (57 + 25)$$

$$12 \times (15 \times 7) = (12 \times 15) \times 7$$

$$35{,}784 + (48{,}12{,}69{,}841 + 3) = (35{,}784 + 48{,}12{,}69{,}841) + 3$$

$$(39{,}458 \times 84{,}321) \times 17 = 39{,}458 \times (84{,}321 \times 17)$$

It is to be noted that here too, subtraction and division are not associative.

1.10.3 Distributivity of multiplication over addition or subtraction

An interesting fact relating to addition and multiplication comes from the following patterns:

$$(72 \times 13) + (28 \times 13) = (72 + 28) \times 13$$

$$37 \times 102 = (37 \times 100) + (37 \times 2)$$

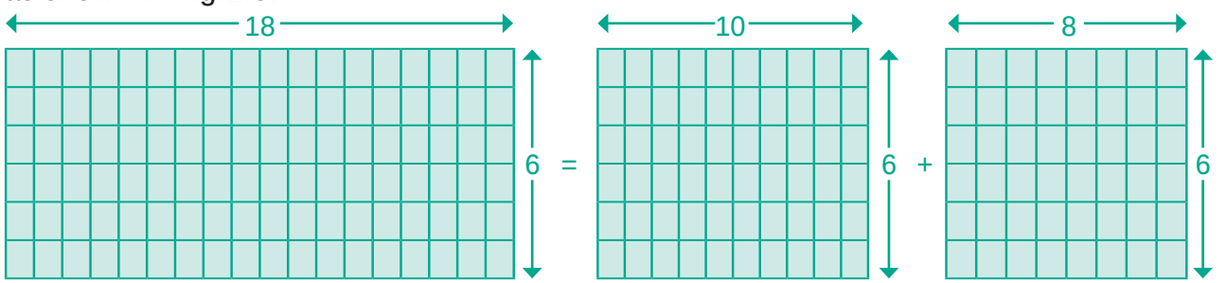
$$37 \times 98 = (37 \times 100) - (37 \times 2)$$

From the last two cases, we are arriving at the following equations:

$$37 \times (100 + 2) = (37 \times 100) + (37 \times 2)$$

$$37 \times (100 - 2) = (37 \times 100) - (37 \times 2)$$

It can be noted that the product of a number and a sum of numbers can be written as the sum of two products. Similarly, the product of a number and a number got by subtraction can be written as the difference of two products. This property is called the property of distributivity of multiplication over addition or subtraction. It is a very useful property to group numbers in a convenient way. Now let us say $18 \times 6 = (10 + 8) \times 6$ in an easy way as shown in Fig.1.3.



(a) (b) (c)

Fig.1.3

Thus, $18 \times 6 = (10 + 8) \times 6$ is shown clearly in the above figure.

It is to be noted that addition does not distribute over multiplication.

For example,

$10 + (10 \times 5) = 60$ and $(10 + 10) \times (10 + 5) = 300$ are not equal.

1.10.4 Identity for addition and multiplication

When zero is added to any number, we get the same number. Similarly, when we multiply any number by 1, we get the same number. So, zero is called the additive identity and one is called the multiplicative identity for whole numbers.

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